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Q1.Load the insurance dataset using pandas and display the first 10 rows of the

data.

import pandas as pd

insurance = pd.read_csv("insurance.csv")

print(insurance.head(10))

Q2.Use pandas functions to get basic information about the dataset:

e Shape of the data (shape) Column names and data types (info() • Statistical summary of numerical columns (describe())

import pandas as pd

insurance = pd.read_csv("insurance.csv")

print("Shape of dataset:")

print(insurance.shape)

print("\nDataset Information:")

insurance.info()

print("\nStatistical Summary:")

print(insurance.describe())

Q3.Check if the dataset contains any null/missing values. Show the count of nul values for each column.

import pandas as pd

```
insurance = pd.read_csv("insurance.csv")
print("Null values in each column:")
print(insurance.isnull().sum())
```

Q4.Identify numerical and categorical columns in the dataset. Print them separately.

```
import pandas as pd
insurance = pd.read_csv("insurance.csv")
numerical_columns = insurance.select_dtypes(include=['int64', 'float64']).columns
categorical_columns = insurance.select_dtypes(include=['object']).columns
print("Numerical Columns:")
print(list(numerical_columns))
print("\nCategorical Columns:")
print(list(categorical_columns))
```

Q5.Create distribution plots for all numerical columns (age, bmi, children, charges):

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
insurance = pd.read_csv("insurance.csv")
plt.figure(figsize=(6,4))
sns.histplot(insurance['age'], kde=True)
plt.title("Distribution of Age")
plt.xlabel("Age")
plt.ylabel("Frequency")
plt.show()
```

```
plt.figure(figsize=(6,4))

sns.histplot(insurance['bmi'], kde=True)

plt.title("Distribution of BMI")

plt.xlabel("BMI")

plt.ylabel("Frequency")

plt.show()

plt.figure(figsize=(6,4))

sns.histplot(insurance['children'], kde=True)

plt.title("Distribution of Number of Children")

plt.xlabel("Children")

plt.ylabel("Frequency")

plt.show()

plt.figure(figsize=(6,4))

sns.histplot(insurance['charges'], kde=True)

plt.title("Distribution of Insurance Charges")

plt.xlabel("Charges")

plt.ylabel("Frequency")

plt.show()
```

Q6.Create count plots for all categorical columns (sex, smoker, region):

Use sns.countplot() for each column

Add titles and labels for clarity

```
import pandas as pd

import seaborn as sns
```

```
import matplotlib.pyplot as plt

insurance = pd.read_csv("insurance.csv")

plt.figure(figsize=(6,4))

sns.countplot(x='sex', data=insurance)

plt.title("Count of Gender")

plt.xlabel("Sex")

plt.ylabel("Count")

plt.show()

plt.figure(figsize=(6,4))

sns.countplot(x='smoker', data=insurance)

plt.title("Count of Smokers and Non-Smokers")

plt.xlabel("Smoker")

plt.ylabel("Count")

plt.show()

plt.figure(figsize=(6,4))

sns.countplot(x='region', data=insurance)

plt.title("Count of Individuals by Region")

plt.xlabel("Region")

plt.ylabel("Count")

plt.show()
```

Q7.Create a heatmap to show the correlation between numerical variables.

• Use sns.heatmap()

Annotate the values and use a suitable color palette

```
import pandas as pd

import seaborn as sns

import matplotlib.pyplot as plt

insurance = pd.read_csv("insurance.csv")

numerical_data = insurance[['age', 'bmi', 'children', 'charges']]

correlation_matrix = numerical_data.corr()

plt.figure(figsize=(8,6))

sns.heatmap(

    correlation_matrix,

    annot=True,

    cmap='coolwarm',

    fmt='.2f',

    linewidths=0.5

)

plt.title("Correlation Heatmap of Numerical Variables")

plt.show()
```

Q8.Perform the following analysis on the charges column:

Find the average insurance charges

Find the maximum and minimum charges • Group by smoker and compare the average charges for smokers vs non- smokers

```
import pandas as pd

insurance = pd.read_csv("insurance.csv")

average_charges = insurance['charges'].mean()

print("Average Insurance Charges:", average_charges)
```

```
maximum_charges = insurance['charges'].max()
print("Maximum Insurance Charges:", maximum_charges)
minimum_charges = insurance['charges'].min()
print("Minimum Insurance Charges:", minimum_charges)
smoker_charges = insurance.groupby('smoker')['charges'].mean()
print("\nAverage Charges by Smoker Status:")
print(smoker_charges)
```

Q9.Create a boxplot to show the distribution of charges with respect to smoker and sex.

(You can create two separate boxplots or use hue parameter)

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
insurance = pd.read_csv("insurance.csv")
plt.figure(figsize=(8,6))
sns.boxplot(x='smoker', y='charges', data=insurance)
plt.title("Distribution of Charges by Smoker Status")
plt.xlabel("Smoker")
plt.ylabel("Insurance Charges")
plt.show()
plt.figure(figsize=(8,6))
sns.boxplot(x='sex', y='charges', data=insurance)
plt.title("Distribution of Charges by Sex")
plt.xlabel("Sex")
```

plt.ylabel("Insurance Charges")

plt.show()

Q10.(Mini Project / Analysis Summary)

Write a short summary (in comments or markdown) answering the following:

1. What is the average age and BMI of the customers?
2. How does smoking affect insurance charges?
3. Which region has the highest number of customers?
4. Any interesting observations from the visualizations?